Q-ORACLE: Hardware-Executed Semantic Temporal Quantum Models for News-Driven Scenario Analysis

Yehor Tereshchenko
Timofey Kruhlov
Nadiia Haidash
Ekaterina Petrova
Mika Hamalainen

QuantumFinances / Q-ORACLE

June 2026

Abstract

Strategic foresight systems built on news streams increasingly use semantic graphs and PESTEL analysis to convert large textual flows into compact descriptions of political, economic, social, technological, environmental and legal change. This paper studies whether such weekly foresight representations can be connected to executable near-term quantum machine learning circuits in a traceable way. We present Q-ORACLE, a pipeline that takes ORACLE-style weekly news graph snapshots, derives PESTEL world-state vectors, constructs 1024-dimensional semantic embeddings from clustered weekly evidence, projects the embeddings into hardware-sized quantum feature maps and samples trained temporal circuits on IQM Sirius quantum hardware. The experimental data adapter consumes the available ORACLE weekly graph archive: 42 dated snapshots from 2025-03-17 to 2026-05-31, containing 1,254 weekly clusters and 17,307 graph links in aggregate. The semantic QML experiment uses `intfloat/multilingual-e5-large-instruct` embeddings, latent circuits of 8, 10 and 12 qubits, two variational layers and a two-bit scenario register. We submit 18 semantic temporal-QML jobs across three seeds, two shot counts and three circuit sizes, obtaining 27,648 hardware shots with no failed submissions. The aggregate hardware distribution is 32.9% stable, 30.9% event-aligned, 18.5% event-divergent and 17.6% uncertain. We do not claim forecasting accuracy or quantum advantage; rather, the contribution is a reproducible architecture and hardware evidence for linking auditable weekly news-graph foresight, semantic temporal representations and quantum circuit execution.

1 Introduction

News-based foresight is useful only when the path from evidence to interpretation remains inspectable. ORACLE-style time-dependent recursive summary graphs (TRSGs) address this by organizing news into weekly clustered graph snapshots and interpreting changes through PESTEL dimensions [1]. Such systems produce compact, auditable world-state representations, but they do not by themselves define a computational experiment for scenario generation.

Q-ORACLE investigates the next step: whether a weekly news-graph foresight state can be transformed into a temporal quantum machine learning (QML) task and executed on real quantum hardware. The question is not whether near-term devices can predict markets. Instead, we ask whether the complete chain can be made concrete: graph snapshots become temporal vectors; temporal vectors become supervised transition targets; targets train shallow variational circuits; and trained circuits are sampled on a quantum processing unit (QPU).

Research question. Given a multi-month sequence of weekly news graph states and an event query, can a semantic temporal representation be encoded into shallow quantum circuits whose measured scenario register produces reproducible hardware-sampled scenario distributions?

Contributions. This paper makes four contributions. First, it defines a temporal scenario task over ORACLE-style weekly graph states and event-conditioned semantic change. Second, it introduces a transparent projection from 1024-dimensional weekly text embeddings to 8–12 qubit quantum feature maps. Third, it reports an 18-job IQM Sirius hardware batch with all job identifiers, counts and aggregate distributions stored in machine-readable artifacts. Fourth, it compares the semantic embedding circuit family against a compact PESTEL-only baseline.

2 Related Work

News foresight and PESTEL. ORACLE converts news streams into weekly TRSGs using embeddings, hierarchical clustering, recursive summarization and PESTEL-aware change detection [1]. PESTEL analysis has also been studied with large language models for operational environment forecasting [2]. Q-ORACLE keeps the PESTEL framing but treats it as a state representation for executable temporal models.

Quantum time-series modeling. Variational quantum circuits have been proposed for one-step time-series forecasting [3], multidimensional generative time-series modeling [4], and long-term multivariate forecasting [5]. However, recent benchmark work cautions that variational quantum approaches often fail to outperform simple classical methods under fair comparison [6]. We therefore evaluate Q-ORACLE as a hardware-backed systems method, not as an advantage claim.

Quantum feature spaces. Quantum machine learning can be viewed as learning in feature Hilbert spaces [7]. At the same time, variational circuits face trainability and noise limitations, including barren plateaus [8]. Q-ORACLE consequently uses shallow ansatzes and explicit dimensionality reduction rather than attempting to map high-dimensional embeddings directly onto physical qubits.

3 Task Definition

Let G_t denote a weekly news graph snapshot at week t, containing clusters, weighted evidence text, graph links and metadata. The foresight system maps each snapshot to a PESTEL vector

$$v_t = [P_t, E_t, S_t, T_t, Env_t, L_t] \in [0,1]^6.$$

For the semantic model, the clustered weekly evidence is embedded into

$$e_t \in \mathbb{R}^{1024},$$

and an event query is embedded into e_{event}. The goal is to learn a temporal circuit that maps the latest weekly representation to a distribution over four scenario classes:

$$\mathcal{C} = \{\text{stable}, \text{event aligned}, \text{event divergent}, \text{uncertain}\}.$$

The output is not interpreted as a calibrated real-world probability. It is a hardware-sampled scenario distribution induced by the specified data, projection, training target and circuit.

4 System Architecture

Figure 1 summarizes Q-ORACLE. The input is a sequence of weekly graph snapshots. The classical side extracts PESTEL features, builds semantic embeddings from cluster text and constructs transition targets. The quantum side encodes projected features into rotations, applies a shallow variational ansatz and measures a two-bit scenario register on hardware.

PESTEL baseline. The compact baseline compresses v_t into four latent variables: political/legal, economic/technological, social/environmental and temporal graph intensity. This path verifies that the PESTEL world-state alone can produce a small circuit and hardware receipt.

Semantic temporal path. The main experiment uses cluster text as the representation source. Cluster embeddings are L2-normalized and averaged using cluster size weights to produce one weekly embedding. This preserves more semantic information than the six PESTEL values while retaining weekly traceability back to source clusters.

5 Semantic Quantum Model

5.1 Embedding Aggregation

For week t, let $c_{t,j}$ be a cluster text and $w_{t,j}$ its cluster-size weight. A sentence embedding model maps each cluster to $\phi(c_{t,j}) \in \mathbb{R}^{1024}$. The weekly semantic state is

$$e_t = \frac{\sum_j w_{t,j}\, \text{norm}(\phi(c_{t,j}))}{\left\| \sum_j w_{t,j}\, \text{norm}(\phi(c_{t,j})) \right\|_2}.$$

The event query is embedded with the same model, yielding e_{event}.

5.2 Projection to Qubit Features

The QPU does not receive 1024 physical qubits. Instead, the embedding is projected into a latent feature vector

$$x_t^{(d)} = \Pi_d(e_t), \qquad d \in \{8, 10, 12\}.$$

The implemented Π_d uses PCA components fitted on the available weekly/event embedding set. When d exceeds the available PCA rank, seeded signed-random projection rows are added. The projected values are scaled through a bounded nonlinearity into $[0,1]$, producing rotation inputs.

5.3 Transition Targets

The semantic target for transition $t \to t+1$ is derived from event alignment and drift. Let

$$s_t = \frac{1 + \cos(e_t, e_{\text{event}})}{2}$$

be the event similarity in $[0,1]$, and let $\Delta_t = s_{t+1} - s_t$. Event-aligned mass increases when $\Delta_t > 0$, event-divergent mass increases when $\Delta_t < 0$, stable mass is penalized by drift, and uncertain mass is increased by week-to-week semantic movement. The resulting four values are normalized to a target distribution y_t.

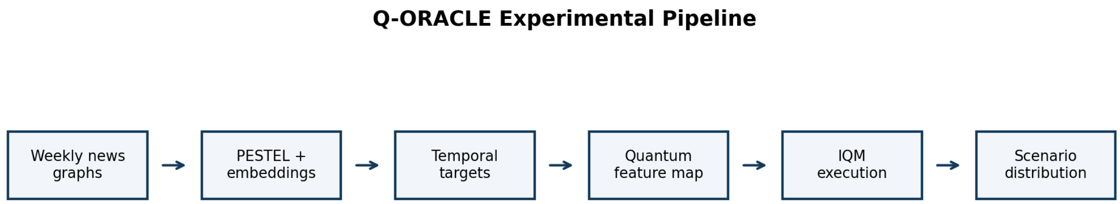

Figure 1: Q-ORACLE pipeline from weekly news graph snapshots to hardware-sampled scenario distributions.

5.4 Circuit and Optimization

For latent dimension d, Q-ORACLE builds a d-qubit circuit. Each feature $x_{t,k}^{(d)}$ is encoded with R_y and R_z rotations. Two variational layers then apply trainable single-qubit rotations and ring entanglement. The first two qubits are measured into a two-bit scenario register. Parameters are trained classically using seeded stochastic search with cross-entropy:

$$\mathcal{L}(\theta) = -\sum_t \sum_{c \in \mathcal{C}} y_{t,c} \log q_{\theta,c}(x_t).$$

Only the trained inference circuit is submitted to the QPU.

6 Experimental Setup

The experiment uses the full available ORACLE graph snapshot archive exposed by the platform adapter: 42 dated weekly graph states spanning 2025-03-17 through 2026-05-31. Together these snapshots contain 1,254 weekly clusters and 17,307 graph links. The latest PESTEL vector is shown in Table 2, and the weekly trajectory is plotted in Figure 2. The semantic embedding model is `intfloat/multilingual-e5-large-instruct`, producing 1024-dimensional vectors. We test latent circuit sizes $d = \{8, 10, 12\}$, seeds $\{41, 42, 43\}$, shot counts $\{1024, 2048\}$, two variational layers and 160 local training iterations.

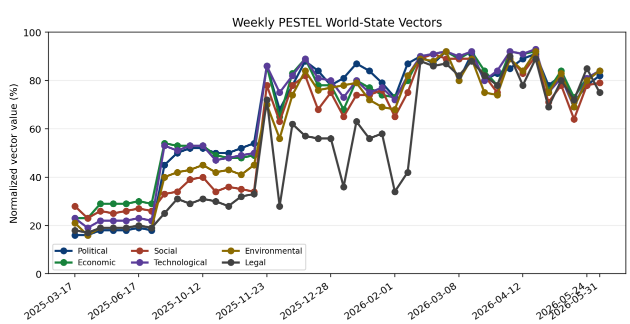

Figure 2: Weekly PESTEL world-state vectors used by the experiment.

All semantic circuits are submitted to IQM Sirius through the Qiskit-on-IQM stack, which exposes IQM Resonance hardware through Qiskit provider and backend interfaces [9]. A run is counted as hardware-backed only when a receipt contains provider, backend, job identifier, circuit fingerprint, shot count and returned counts.

7 Results

Class	Local QML	IQM hardware
Stable	49.4%	41.4%
Aligned Up	26.2%	20.3%
Aligned Down	9.4%	25.0%
Uncertain	15.0%	13.3%

Table 1: Temporal QML scenario distribution. The local column is the statevector prediction after classical training; the hardware column is sampled from IQM Sirius.

Dimension	Latest value
Political	82.0%
Economic	84.0%
Social	79.0%
Technological	84.0%
Environmental	84.0%
Legal	75.0%

Table 2: Latest weekly PESTEL vector used for feature encoding.

Semantic model	E5-large-instruct
Embedding dimension	1024
Hardware jobs	18
Total hardware shots	27648
Latent qubits tested	8, 10, 12

Table 3: Semantic-embedding QML batch configuration.

Q	N	Shots	Stable	Align	Diverg.	Unc.
8	6	9216	31.1%	30.1%	20.0%	18.9%
10	6	9216	34.0%	31.1%	18.3%	16.5%
12	6	9216	33.7%	31.5%	17.3%	17.5%

Table 4: Mean IQM Sirius distributions for the 1024-dimensional semantic-embedding QML batch.

7.1 Compact PESTEL Baseline

The compact four-qubit temporal circuit was submitted once with 128 shots. The hardware sample distribution differs from the local statevector estimate, especially for the aligned-down branch. This is expected for a shallow

Class	Overall mean
Stable	32.9%
Event Aligned	30.9%
Event Divergent	18.5%
Uncertain	17.6%

Table 5: Aggregate distribution across all semantic-embedding QML hardware jobs.

circuit on noisy hardware with a small shot budget and is reported as a baseline rather than a calibrated event estimate.

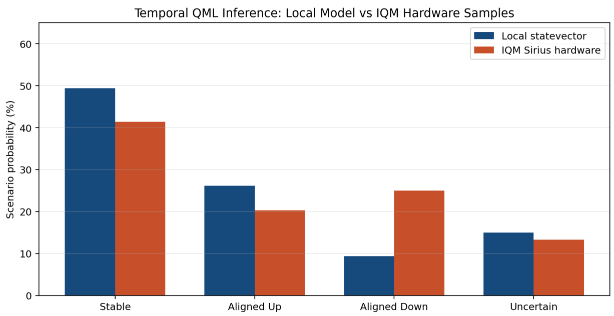

Figure 3: Compact PESTEL QML baseline: local statevector distribution compared with IQM Sirius samples.

7.2 Semantic Embedding Hardware Batch

The semantic experiment submits 18 jobs: three latent sizes, three seeds and two shot counts. Table 4 summarizes mean hardware distributions by circuit size, while Table 5 gives the aggregate distribution. Across all semantic jobs, the largest branch is stable at 32.9%, followed closely by event-aligned at 30.9%. The 12-qubit setting has the highest event-aligned mean among the tested configurations.

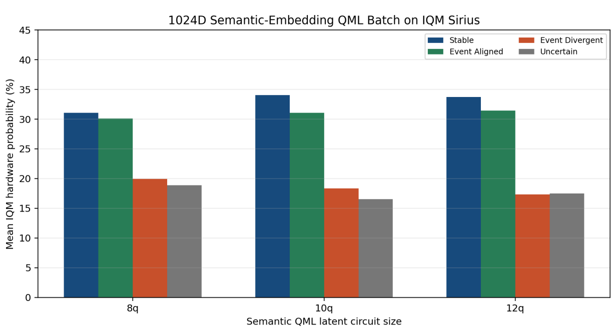

Figure 4: Mean IQM Sirius distributions for 1024-dimensional semantic embeddings projected into 8-, 10- and 12-qubit circuits.

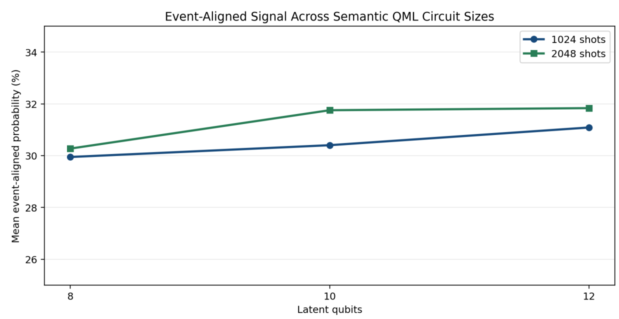

Figure 5: Event-aligned probability by latent circuit size and shot count.

7.3 Operational Evidence

The experimental trail is represented by two artifact layers. The numerical layer is the set of JSON receipts and batch tables used above. The operational layer is shown in Figures 6 and 7: the local scenario interface renders the complete PESTEL-to-scenario path, and IQM Resonance records the completed hardware jobs on IQM Sirius.

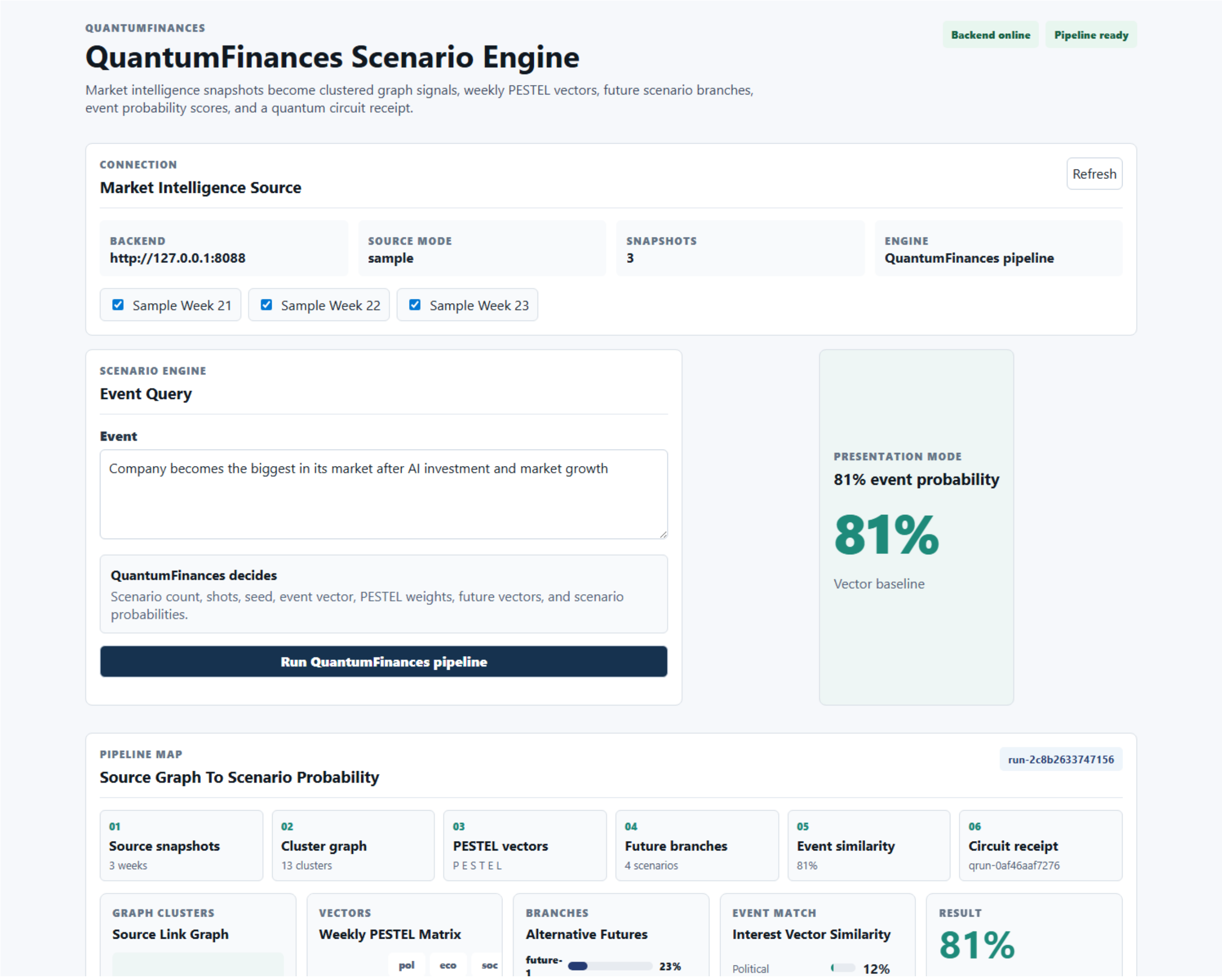

Figure 6: QuantumFinances scenario interface after a completed run, showing source snapshots, PESTEL vectors, future branches, event scoring and circuit receipt in one audit view.

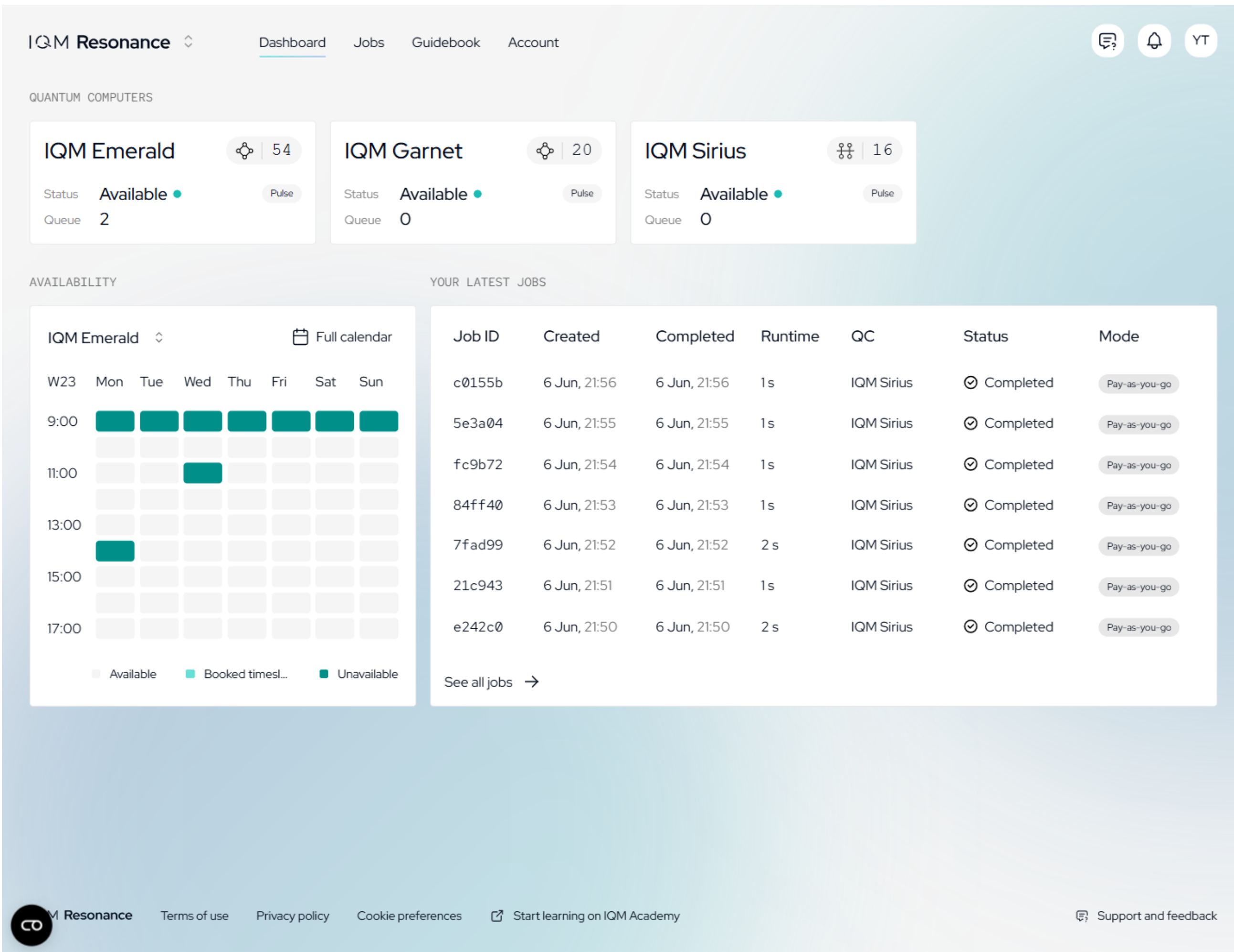

Figure 7: IQM Resonance dashboard after the hardware batch, showing available quantum computers and completed IQM Sirius jobs corresponding to the submitted experiment sequence.

8 Discussion

The results show that an auditable news-graph state can be carried through a semantic temporal representation, projected into shallow quantum feature maps and executed repeatedly on a real QPU. The strongest empirical signal is methodological: all 18 semantic jobs completed successfully, and the generated artifacts preserve a trace from input run to embeddings, circuit parameters, provider job identifiers and measured counts.

The hardware distributions should not be read as market predictions. They are scenario signals produced by the chosen semantic target and circuit family. The value of the experiment is that these signals are reproducible and hardware-grounded rather than hand-written or simulated receipts. The comparison between 8-, 10- and 12-qubit circuits also illustrates that the embedding projection and circuit size affect the measured distribution, which must be accounted for in any future validation study.

9 Limitations

The current ORACLE archive used here is larger than the initial prototype but still limited for statistical forecasting validation: 42 weekly snapshots provide 41 adjacent transition examples. Q-ORACLE consumes the full dated snapshot set exposed to the adapter, so the remaining limitation is archive length and coverage rather than a selected subset. The 1024-dimensional embeddings are real model outputs, yet the projection to 8–12 qubits is an information bottleneck. The ansatz is shallow to remain hardware-feasible; larger circuits may be more expressive but will also be harder to optimize and noisier. Finally, Q-ORACLE produces event-conditioned scenario distributions, not investment advice or verified future probabilities.

10 Conclusion

Q-ORACLE connects ORACLE-style weekly news graph foresight to hardware-executed temporal QML. The system derives PESTEL and semantic weekly states from a 42-week archive, projects 1024-dimensional embeddings into small quantum feature maps, trains shallow temporal circuits and samples scenario registers on IQM Sirius. The present study contributes a reproducible method and hardware evidence for linking narrative intelligence, semantic vectors and near-term quantum processors. Future work should extend the ORACLE archive, compare against classical temporal baselines and evaluate whether hardware-sampled scenario distributions add value beyond transparent classical models.

References

[1] L. Kharlashkin, E. Morooka, Y. Tereshchenko, and M. Hamalainen. ORACLE: Time-Dependent Recursive Summary Graphs for Foresight on News Data Using LLMs. arXiv:2512.15397, 2025.

[2] K. Alnajjar and M. Hamalainen. MLPestel: The New Era of Forecasting Change in the Operational Environment of Businesses Using LLMs. Centria University of Applied Sciences, 2024.

[3] P. Kaushik, S. Pramanik, M. G. Chandra, and C. V. Sridhar. One-Step Time Series Forecasting Using Variational Quantum Circuits. arXiv:2207.07982, 2022.

[4] H. Horowitz, P. Rao, and S. K. Radha. A Quantum Generative Model for Multi-Dimensional Time Series Using Hamiltonian Learning. arXiv:2204.06150, 2022.

[5] H. H. S. Chittoor, P. R. Griffin, A. Neufeld, J. Thompson, and M. Gu. QuLTSF: Long-Term Time Series Forecasting with Quantum Machine Learning. arXiv:2412.13769, 2024.

[6] T. Fellner, D. Kreplin, S. Tovey, and C. Holm. Quantum vs. Classical: A Comprehensive Benchmark Study for Predicting Time Series with Variational Quantum Machine Learning. arXiv:2504.12416, 2025.

[7] M. Schuld and N. Killoran. Quantum Machine Learning in Feature Hilbert Spaces. Physical Review Letters, 122:040504, 2019.

[8] M. Larocca et al. A Review of Barren Plateaus in Variational Quantum Computing. arXiv:2405.00781, 2024.

[9] IQM. Qiskit on IQM User Guide. IQM Client Documentation, 2026. https://docs.iqm.tech/iqm-client/user_guide_qiskit.html.